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Workforce Upskilling in the AI-Driven Knowledge Economy: A Systematic Review of Research Trends and Organizational Learning Strategies

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Abstract: Artificial intelligence (AI) and automation are reshaping skill requirements in the knowledge economy, placing increasing pressure on organizations to reconsider how employees acquire and update their competencies. Although research on AI-related workforce development continues to expand, the relationships among technological skills, organizational learning, and sector-specific upskilling strategies remain insufficiently integrated across disciplines. This study examines recent research trends in workforce upskilling and identifies the principal competencies and organizational responses associated with AI-driven transformation. A systematic literature review was conducted using EBSCO Discovery Service (a cross-database discovery platform), covering peer-reviewed publications from January 2022 to December 2025. Publication distributions, keyword frequencies, and recurring research themes were examined to characterize the development of the field. The analysis identified five interconnected areas: foundational digital literacy, AI-specific technical competencies, human-centered skills, organizational learning systems, and sector-specific adaptation strategies. The reviewed literature also showed a growing emphasis on continuous learning, AI-supported training, and the integration of skill development into organizational practices, although approaches differed across industries and institutional contexts. These findings indicate that workforce adaptation to AI depends on the interaction between individual competencies and organizational learning arrangements rather than technical training alone. By bringing together research from management, education, economics, and related fields, this review provides an integrated account of upskilling priorities and identifies directions for research on knowledge development, organizational adaptation, and workforce transformation.

Keywords: Artificial intelligence; Knowledge economy; Workforce upskilling; Organizational learning; Digital competencies; Workforce transformation; Systematic literature review

1 Introduction

Artificial intelligence (AI), automation, generative AI, and data-driven technologies are changing how work is organized, performed, and evaluated across the global economy. Organizations increasingly incorporate intelligent systems into operational processes, decision-making, customer service, and knowledge-intensive activities [1]. These developments have brought workforce capabilities into sharper focus, as access to advanced technologies alone does not determine how effectively organizations can use them. The ability of employees to understand, apply, and adapt to new technologies has become an important consideration in organizational competitiveness [2]. Consequently, workforce development is receiving greater attention from businesses, industry representatives, and policymakers seeking to address the changing demands of technology-intensive work [3–5].

These changes are particularly relevant to the knowledge economy, where economic value depends increasingly on the creation, application, and exchange of knowledge, together with the skills and learning processes that support innovation [6] . AI introduces new possibilities for productivity, organizational efficiency, and innovation, but its adoption also alters occupational structures and the competencies required to perform existing and emerging tasks [7, 8]. Some activities are being automated, while others require closer interaction between employees and intelligent

systems. Workers must therefore update their knowledge and skills throughout their careers, and organizations face growing pressure to reconsider established approaches to employee training, professional development, and talent management [9–11]. From a knowledge management perspective, this transition raises questions about how individual learning is translated into organizational capabilities and how newly acquired competencies are shared and applied in everyday work.

Employee training has traditionally been organized around periodic activities intended to address immediate operational requirements [12]. Such arrangements are increasingly difficult to reconcile with the pace and breadth of technological change, particularly when employees must repeatedly adjust to unfamiliar tools, processes, and responsibilities [13]. Continuous learning, upskilling, and reskilling have consequently become central concerns in workforce development. Although these concepts are closely related, they describe different forms of professional adjustment. Upskilling refers to the development of additional or more advanced competencies within an existing occupational role, whereas reskilling involves preparing individuals to undertake substantially different tasks or move into new roles [14]. Both require more than the provision of isolated training courses. Their implementation also depends on organizational arrangements that allow employees to acquire, exchange, and apply knowledge as working practices evolve.

The period beginning in 2022 is particularly relevant to these developments. The wider availability of generative AI applications, advances in machine learning, and the continuing expansion of automation have accelerated discussions about the future composition of workforce skills [15, 16]. Whereas earlier phases of automation were frequently associated with routine and repetitive activities, recent AI applications increasingly affect professional judgment, analytical work, communication, and creative tasks [17, 18]. The implications extend beyond technical occupations to management, education, healthcare, manufacturing, and other knowledge-intensive settings. Accordingly, workforce development is no longer concerned solely with teaching employees to operate new technologies. It also involves identifying which human capabilities remain important, how technical and nontechnical competencies should be combined, and how learning opportunities can be maintained across different organizational and sectoral contexts.

Research addressing these questions has expanded across management, education, economics, labor studies, engineering, information systems, and public policy. However, the literature remains dispersed across disciplinary perspectives that frequently examine different aspects of the same transformation. Studies concerned with employability and labor market adjustment often emphasize changing occupational requirements, while research on digital competencies focuses more directly on individual knowledge and technical proficiency. Organizational studies, meanwhile, consider employee development in relation to learning practices, management decisions, and institutional arrangements. These perspectives are closely connected, yet they are not always examined together. Consequently, the relationship between individual upskilling, organizational learning, and broader workforce transformation remains insufficiently integrated in existing accounts of the AI-driven knowledge economy.

A further difficulty concerns the diversity of skills and learning arrangements discussed in recent research. Digital literacy, AI-specific technical competencies, critical thinking, communication, and other human-centered capabilities are frequently addressed alongside organizational training systems and industry-specific adaptation strategies. However, identifying these competencies individually does not fully explain how they relate to one another or how organizations incorporate them into continuing workforce development. A synthesis that considers both the distribution of recent research and its principal themes is therefore needed to clarify the connections between technological change, knowledge development, and organizational responses. Such an examination is particularly relevant to knowledge and innovation studies, where the creation and application of knowledge depend on the interaction between individual capabilities, institutional practices, and changing technological conditions.

Against this background, this study examines workforce upskilling in the AI-driven knowledge economy through a systematic review of peer-reviewed academic literature published between January 2022 and December 2025. The selected period captures the recent expansion of generative AI and related technologies in organizational and economic settings. Drawing on records retrieved through the EBSCO Discovery Service (a cross-database discovery platform), the review examines publication patterns, keyword frequencies, and recurring themes in research on AI-related workforce development. Particular attention is given to the competencies identified in the literature, the movement toward continuous and technology-supported learning, and differences in organizational responses across sectors. Rather than treating upskilling as an isolated human resource activity, the study considers its relationship with organizational learning and the development of knowledge-based capabilities. Its contribution lies in bringing together research from several disciplinary areas to provide a structured account of recent developments and to identify questions that require further investigation.

Five research questions guide the review:

[RQ1:] How have workforce upskilling strategies evolved in the AI-driven knowledge economy?

[RQ2:] What are the dominant research themes in recent studies of AI-related workforce development?

[RQ3:] Which digital, AI-specific technical, and human-centered competencies receive the greatest attention in

the literature?

[RQ4:] How are organizations adapting conventional training practices to support continuous learning and the development of workforce capabilities?

[RQ5:] What limitations in the existing literature and directions for future research can be identified from the reviewed studies?

The remainder of the paper is organized as follows. Section 2 describes the review methodology, literature search, and study selection procedures. Section 3 presents the descriptive results, including publication distributions across publishers, journals, and countries, together with keyword frequency analysis. Section 4 discusses the principal research themes, their implications for workforce development and organizational learning, and the remaining research gaps and methodological limitations. Section 5 summarizes the main conclusions and their relevance to workforce upskilling in the AI-driven knowledge economy.

2 Materials and Methods

This study adopted a systematic literature review approach informed by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [19]. The review examined research on upskilling, reskilling, and workforce transformation in the context of AI and the changing knowledge economy. Particular attention was given to the competencies required in AI-related work, the development of organizational learning practices, and the ways in which institutions and industries respond to changing skill requirements. The EBSCO Discovery Service was used as the primary literature retrieval platform because it provides access to scholarly publications across management, education, economics, information systems, and related disciplines. The general search and review procedure is presented in Figure 1.

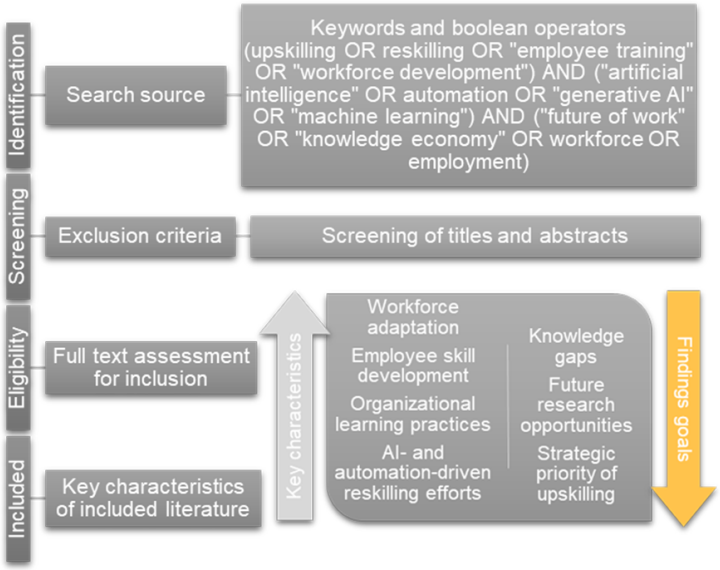


Figure 1. Literature search and review procedure

The literature search combined three groups of terms covering workforce development, AI-related technologies, and changes in employment and the knowledge economy. Boolean operators were used to identify publications addressing these areas in combination. The search covered full-text content and available bibliographic fields, including titles, abstracts, keywords, and subject descriptors, according to the search options provided by EBSCO Discovery Service. The search expression was as follows:

("upskilling" OR "reskilling" OR "employee training" OR "workforce development") AND ("AI" OR "automation" OR "generative AI" OR "machine learning") AND ("future of work" OR "knowledge economy" OR "workforce" OR "employment")

The initial search returned 158,090 records. When the results were restricted to academic journals, 32,068 records remained. These figures represented search results rather than a set of unique, eligible studies, as records indexed through more than one database could appear multiple times. The distinction between retrieved records and studies meeting the review criteria was therefore important when interpreting the size and coverage of the literature.

The review was restricted to peer-reviewed journal articles published in English between January 2022 and December 2025. This period was selected to examine recent developments in workforce skills following the wider introduction of generative AI applications, the growing use of machine learning in organizational settings, and

the continued adoption of automation and digital technologies. It also provided a defined period for examining how research attention had shifted toward continuous learning, workforce adaptation, and the changing relationship between technological capabilities and human expertise. The purpose of this restriction was to concentrate on contemporary research rather than to provide a historical account of workforce development.

The inclusion criteria focused on studies that addressed upskilling, reskilling, employee development, or workforce transformation in relation to AI and automation. Publications were considered relevant when they examined changes in occupational competencies, organizational learning and talent development, employment-related adjustment, or institutional responses to technological change. Studies concerned with knowledge acquisition, professional learning, and the application of new competencies in organizational settings were also within the scope of the review. These criteria reflected the study's interest in workforce upskilling as both an individual learning process and an aspect of organizational adaptation in the knowledge economy.

Publications were excluded when their primary concern was the technical development of AI systems without a substantive connection to workforce skills or employment. Studies that did not address workforce development, upskilling, or reskilling were also excluded, as were publications outside the specified period, non-English publications, and sources that did not meet the peer-review requirement. Research with no identifiable relationship to organizational practices, labor-market change, or the development of workforce capabilities was considered outside the scope of the review.

The screening procedure began with an examination of titles and abstracts to determine whether the retrieved publications addressed the research objectives. Records that did not meet the eligibility criteria were excluded from further consideration. Full-text articles were then examined for their relevance to AI-related workforce development and the knowledge economy. The assessment considered whether each study addressed one or more of the following areas: adaptation to AI and automation, employee skills and lifelong learning, organizational learning and talent management, upskilling and reskilling practices, labor-market transformation, and the implications of AI adoption for knowledge-intensive work.

The review examined the distribution of relevant publications across publishers, journals, and countries, together with the frequency of keywords associated with workforce development and technological change. These descriptive analyses were used to characterize the research landscape and identify recurring areas of academic attention. The subsequent discussion considered the principal competencies and organizational responses represented in the literature, with particular attention to digital literacy, AI-specific technical skills, human-centered capabilities, continuous organizational learning, and differences in upskilling requirements across sectors. The interpretation focused on relationships among these areas and their relevance to knowledge development and workforce adaptation rather than on estimating the causal effects of individual training interventions.

3 Primary Results

The database search identified 17,827 records of peer-reviewed journal articles published between January 2022 and December 2025 that met the specified search restrictions, including 13,382 records with full-text access. These results indicated substantial research activity concerning workforce development, technological change, and organizational adaptation during the four-year period. The volume of retrieved literature also reflected the increasing attention given to upskilling and reskilling as organizations responded to the adoption of AI and related technologies. However, the retrieved records should be distinguished from a fully screened set of unique eligible studies, as the search results may include overlapping database entries and publications with varying degrees of relevance.

The retrieved literature covered management, economics, education, engineering, information systems, public policy, sustainability studies, labor economics, and organizational psychology. This disciplinary range suggested that AI-related workforce transformation was being examined beyond conventional human resource management, particularly in relation to organizational knowledge, technological adaptation, and institutional responses to changing employment requirements. It also pointed to the need for examining individual competencies alongside the organizational and socioeconomic conditions under which those competencies are developed and applied [8, 20].

3.1 Publisher Distribution

Approximately 100 publishers were represented in the retrieved records, although publication activity was concentrated among a smaller group of major academic publishers. MDPI accounted for the largest number of publications, with 4,208 records, followed by Springer with 1,159 and Wiley-Blackwell with 1,031. Taylor & Francis contributed 796 records, while Emerald Publishing accounted for 640. Other publishers represented in the dataset included Frontiers Media (497), Sage (388), the Public Library of Science (293), and BioMed Central (228). These figures described the distribution of records returned by the search rather than a direct measure of the scholarly influence or quality of individual publishers.

The distribution shown in Figure 2 indicated that research associated with AI-related upskilling appeared across a range of multidisciplinary and subject-specific publishing outlets. The comparatively large representation of

MDPI was consistent with the presence of journals covering sustainability, management, technology, and education. Emerald Publishing's contribution reflected the inclusion of research concerned with business management, organizational practices, and workforce development. The representation of Springer, Taylor & Francis, Wiley-Blackwell, and Sage further illustrated the breadth of publishing activity across established disciplinary fields. Taken together, these patterns suggested that workforce upskilling had become a subject of interest across several areas of scholarship rather than remaining confined to a single research community.

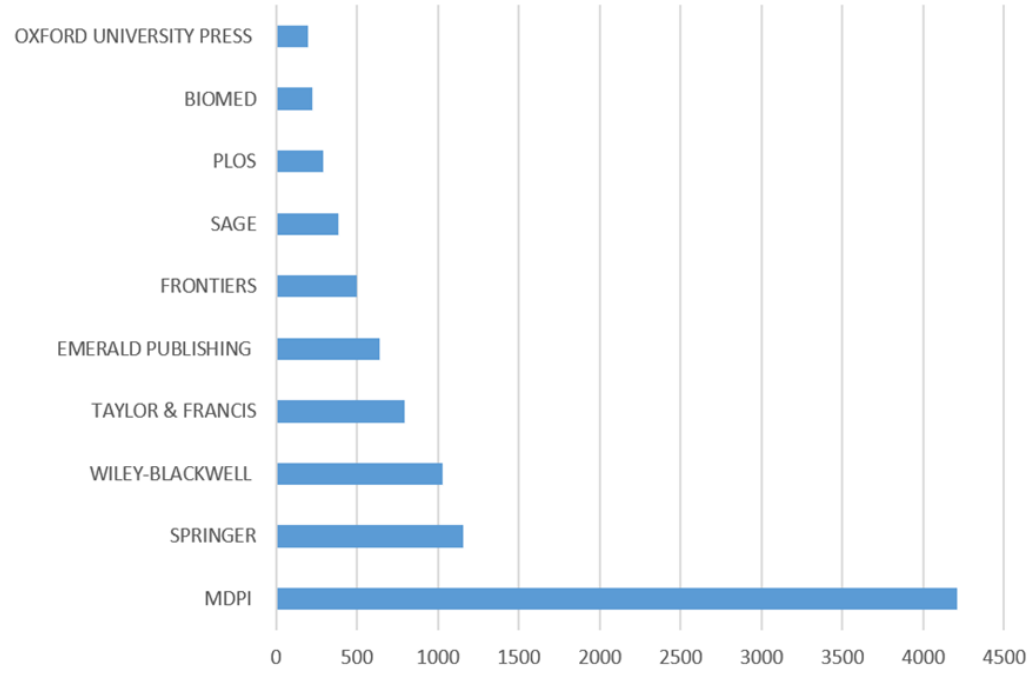


Figure 2. Distribution of publications in AI-related upskilling research

Note: AI = artificial intelligence.

3.2 Journal Distribution

Approximately 300 academic journals were represented in the retrieved literature, reflecting the range of disciplines in which AI-related workforce development was discussed. Publication counts nevertheless varied considerably across journals (Figure 3). Sustainability accounted for the largest number of records, with 1,980 publications, followed by PLOS ONE (278), Systems (159), Economies (121), Frontiers in Public Health (113), Administrative Sciences (110), Applied Sciences (91), and Education Sciences (86). The concentration of records in a limited number of journals indicated that certain interdisciplinary outlets were particularly prominent within the search results.

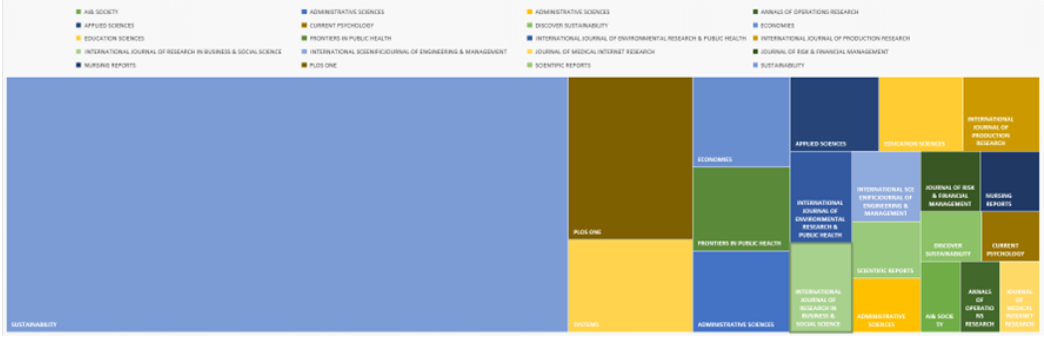


Figure 3. Top journals publishing research on upskilling and AI-driven workforce transformation

The representation of Sustainability was consistent with research examining organizational resilience, responsible innovation, social inclusion, and labor-market transitions in the context of technological change [21–25]. Publications in Systems reflected an interest in the relationships among technological systems, organizational processes, and

workforce requirements [26–28]. Research appearing in Economies [29–31] and Administrative Sciences [32–34] addressed related questions concerning governance, economic activity, managerial practices, and organizational adjustment. Meanwhile, the presence of Education Sciences [35–37] pointed to the role of educational institutions, professional development, and lifelong learning in preparing individuals for changing skill requirements. These publication patterns suggested that the study of upskilling involved both the development of individual competencies and the institutional arrangements through which knowledge and skills are acquired.

3.3 Leading Nations

Country-level publication patterns were examined using records for which author-affiliation information was available. Among the countries reported in the dataset, China accounted for the largest number of publications (288), followed by the United States (247), the United Kingdom (147), Australia (146), and India (145). The distribution indicated that research activity was particularly visible in countries with substantial higher education, technological, and industrial research systems. At the same time, these counts represented the available affiliation data and should not be interpreted as a complete measure of national research output or the extent of workforce upskilling within each country.

Several contextual factors may help explain the observed distribution. China's research activity occurred against a background of investment in AI, digital manufacturing, and technology-oriented industrial development. In the United States, research on workforce transformation was closely relevant to AI adoption, higher education, entrepreneurship, and changing employment practices. India's representation was also notable given the scale of its service economy, digital workforce, and continuing interest in employability and professional skills. Research from the United Kingdom and Australia was similarly relevant to discussions of lifelong learning, future workforce requirements, and digital competitiveness [38–41]. Although these national circumstances provided useful context, the publication counts alone did not establish which factors accounted for differences between countries.

3.4 Keyword Analysis

Keyword frequency analysis was used to examine the terminology associated with AI-related workforce development and to identify prominent subjects within the retrieved literature. The most frequently recorded term was "artificial intelligence", with 2,811 occurrences, followed by "digital transformation" (1,054), "sustainability" (989), "digital technology" (982), "technological innovations" (853), and "industry 4.0" (813). Other frequently occurring terms included "sustainable development" (578), "personnel management" (506), "environmental issues" (472), "machine learning" (456), "company business management" (450), "labor market" (419), "career development" (400), "automation" (377), "labor supply" (325), and "professional employee training" (277).

The distribution presented in Figure 4 and Table 1 showed that the retrieved literature addressed several related areas, including technological adoption, employee competencies, organizational practices, and labor-market change. Terms associated with AI and digital technologies occurred alongside those concerning human capital, professional development, management, and employment. This pattern suggested that workforce upskilling was discussed in connection with both technological implementation and the organizational conditions surrounding its use. However, keyword frequencies alone did not establish the strength of relationships between research topics or the existence of distinct thematic clusters.

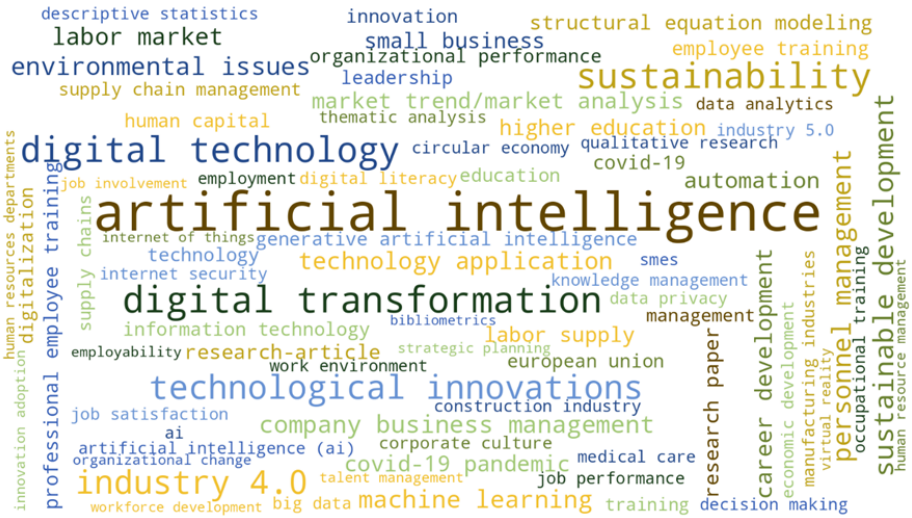


Figure 4. Keyword frequency distribution

Table 1. Keyword frequency distribution

REP	TERM	REP	TERM	REP	TERM	REP	TERM
2811	Artificial intelligence	304	Research-article	201	Corporate culture	152	Economic development
1054	Digital transformation	304	Structural equation modeling	191	Employment	150	Data privacy
989	Sustainability	303	Research paper	190	Artificial intelligence (AI)	149	Medical care
982	Digital technology	277	Professional employee training	185	Decision making	149	Occupational training
853	Technological innovations	272	Employee training	185	Thematic analysis	148	SMEs
813	Industry 4.0	270	COVID-19	182	AI	144	Organizational change
578	Sustainable development	262	Information technology	180	Job satisfaction	143	Human resource management
506	Personnel management	259	Human capital	176	Industry 5.0	143	Human resources departments
472	Environmental issues	258	Digitalization	173	Circular economy	143	Strategic planning
456	Machine learning	251	Innovation	173	Descriptive statistics	142	Internet of things
450	Company business management	251	Technology	171	Data analytics	141	Employability
419	Labor market	245	European Union	166	Knowledge management	141	Job involvement
412	Technology application	243	Education	166	Qualitative research	140	Innovation adoption
400	Career development	241	Generative AI	165	Construction industry	140	Talent management
377	Automation	241	Leadership	162	Big data	135	Workforce development
358	COVID-19 pandemic	237	Organizational performance	162	Work environment	131	Virtual reality
355	Market trend/market analysis	224	Management	160	Manufacturing industries	130	Bibliometrics
343	Higher education	211	Supply chains	159	Digital literacy	123	Labor productivity
343	Small business	208	Supply chain management	157	Internet security	116	Employability
325	Labor supply	208	Training	156	Job performance	...	-

Note: REP = Repetitions (frequency of keyword occurrences); AI = Artificial intelligence; SMEs = Small and medium-sized enterprises; Keyword frequencies are reported as retrieved from the original database records. Variations in terminology, including abbreviations and alternative forms of the same concept, were retained without consolidation, The ellipsis (...) indicates that additional keywords with lower frequencies are omitted for brevity; the dash (-) indicates no further data.

Technological adoption and digital innovation constituted a prominent area of discussion. Terms such as "AI," "artificial intelligence," "machine learning," "big data," "automation," "Internet of Things," "data analytics," "digitalization," and "technology application" reflected research concerned with the introduction and use of advanced technologies in organizational activities [42–44]. The appearance of "generative AI" was particularly relevant to the review period, as recent applications of language models and other generative systems have introduced new questions about the competencies required for knowledge-intensive work [45]. These terms pointed to a research interest extending beyond the technical characteristics of AI toward its use in professional and organizational settings.

A second area concerned workforce competencies and professional development. Terms including "digital literacy," "talent management," "employee training," "professional development," "occupational training," "career growth," "employability," "human capital," and "training" indicated attention to the acquisition and maintenance of workforce capabilities in changing technological environments [11]. Their presence was consistent with research examining how organizations prepare employees to work with AI systems and respond to altered occupational requirements. Within this literature, workforce development was considered not only in relation to technology adoption but also in connection with the capacity of organizations to retrain employees and support movement into changing roles [46, 47].

Organizational adaptation formed another important area of the retrieved literature. Terms such as "organizational performance," "leadership," "knowledge management," "strategic planning," "decision making," "corporate culture," "organizational change," "management," and "job performance" reflected the attention given to internal organizational processes and the management of technological change [48]. These subjects were relevant to understanding how individual learning could be connected to organizational knowledge and working practices. The literature therefore placed workforce development within a broader discussion of management decisions, organizational culture, and the arrangements through which newly acquired skills are applied in practice [49].

Sector-related terminology also appeared frequently. Keywords such as "supply chains," "supply chain management," "manufacturing industries," "construction industry," "Industry 4.0," and "Industry 5.0" indicated an interest in industries undergoing technological and operational change. In these settings, the introduction of automation, digital systems, and increasingly interconnected production processes raised questions about the competencies required of employees and the organization of workplace learning. The prominence of these terms was consistent with research on reskilling in manufacturing, logistics, construction, and other operationally intensive sectors [50].

The occurrence of "sustainability," "circular economy," "sustainable development," "environmental issues," and "economic development" indicated that workforce development was also discussed in relation to wider economic and societal objectives. These subjects connected technological change with questions of responsible organizational practice, resource use, employment conditions, and the distribution of opportunities to acquire new skills. Within

this context, upskilling was relevant not only to organizational adjustment but also to the ability of workers and institutions to participate in economic and technological transitions [51–53].

A further group of terms concerned employment and institutional responses to workforce change. Keywords including "employment," "labor market," "labor supply," "European Union," "higher education," and "education" reflected research on the relationship between skills, employment opportunities, and public or educational institutions. The associated literature addressed matters such as education reform, employability initiatives, vocational preparation, and labor-market measures intended to respond to changing occupational requirements. These subjects placed workforce upskilling within the wider relationship between knowledge development, institutional policy, and employment adjustment [54–58].

Other frequently recorded terms pointed to issues associated with particular technologies and changes in working arrangements. "Data privacy" was relevant to concerns about information governance, employee monitoring, and the responsible use of AI in organizational settings. "Virtual reality" reflected interest in immersive technologies and their possible applications in professional training. The occurrence of "COVID-19" and "COVID-19 pandemic" also indicated that part of the retrieved literature examined workforce development in relation to remote work, changing organizational routines, and the wider adoption of digital practices following the pandemic [59–63].

Methodological terminology provided an additional indication of the approaches represented in the retrieved literature. Terms such as "structural equation modeling," "thematic analysis," "bibliometrics," "qualitative research," and "descriptive statistics" suggested that researchers had examined AI-related workforce development using different analytical perspectives. These approaches addressed questions ranging from relationships among organizational variables to the interpretation of employee experiences and the mapping of publication trends. Their presence illustrated methodological diversity, although the keyword counts did not establish how frequently each method was actually applied or whether the methods were used in combination [64–68].

Overall, the publication distributions and keyword frequencies showed that research on workforce upskilling in the AI-driven knowledge economy extended across technological, educational, organizational, and socioeconomic concerns. The retrieved literature repeatedly addressed the relationship between AI adoption and changing competency requirements, while also drawing attention to leadership, organizational learning, and the institutional conditions surrounding workforce development. The descriptive findings were consistent with a movement in research attention from isolated training activities toward more continuous approaches to skills development. They also provided a basis for examining, in the following section, how digital literacy, AI-specific technical competencies, human-centered capabilities, organizational learning systems, and sector-specific adaptation strategies were discussed across the reviewed literature.

4 Discussion on Review Results

Workforce upskilling has become an important concern in human resource development as AI changes the competencies required in knowledge-intensive work [69, 70]. The literature also places skill development within broader efforts by organizations, industries, and governments to respond to technological change [11]. The review of 13,382 full-text records published between 2022 and 2025 pointed to growing attention to approaches that connect individual competencies with organizational learning and workforce adaptation. Rather than treating training as a separate personnel activity, recent research increasingly considers how employees acquire, apply, and update knowledge as AI becomes part of established working practices.

The findings brought together research from several disciplines, particularly management, education, economics, information systems, and public policy. Three concerns appeared repeatedly: technological change, workforce capability development, and organizational adaptation. Related discussions addressed sustainability, industrial modernization, labor-market restructuring, and digital inclusion. Taken together, these areas suggested that workforce adaptation involves more than the introduction of new technologies. It also depends on the learning arrangements, institutional conditions, and individual competencies that determine how technologies are used in practice. Five interconnected areas provided a useful basis for interpreting the reviewed literature: foundational digital literacy, AI-specific technical competencies, human-centered capabilities, organizational learning systems, and sector-specific adaptation strategies. These areas should not be understood as successive stages that every employee or organization must follow. Instead, they represent related dimensions of workforce development whose importance varies according to occupational responsibilities, technological requirements, and organizational circumstances.

The first area concerns foundational digital literacy. Research on AI-related skills frequently identifies basic digital competencies as an important condition for participating in technology-intensive work [71–73]. These competencies include the ability to use digital systems, interpret information, recognize cybersecurity risks, and acquire unfamiliar technical knowledge. Digital literacy also involves the confidence and judgment needed to work in environments where information is increasingly generated, processed, and communicated through digital platforms [74–78]. Its significance extends beyond routine computer use because employees must be able to evaluate information, recognize the limitations of digital tools, and adjust their working practices as technologies change. Workers with limited digital

experience may encounter greater difficulties when more specialized AI training is introduced. From this perspective, digital literacy is closely related to access to learning opportunities and participation in changing labor markets [79]. Organizations therefore need to consider differences in employees' existing competencies when developing training arrangements, rather than assuming that all workers can begin AI-related learning from the same level.

The second area involves AI-specific technical competencies. These extend beyond general digital literacy to include the use of AI applications, the interpretation of algorithmic outputs, and an understanding of how intelligent systems operate in particular occupational settings. Prompt engineering has received increasing attention as employees interact with generative AI applications through natural-language instructions [80, 81]. Other studies examine the use of AI tools in professional activities [82–84], knowledge of machine learning concepts [85, 86], and the application of data analysis and automated workflows [87]. The relevance of these competencies is not limited to occupations traditionally associated with information technology. Employees in finance, marketing, administration, and operational management increasingly encounter AI-supported systems in routine and complex tasks. Recent research also considers how workers assess generated information, recognize errors, and incorporate algorithmic recommendations into professional judgment [88–92]. This suggests that AI-related competence cannot be reduced to the ability to operate particular software applications. It also involves understanding when AI outputs are appropriate, how they should be interpreted, and where human assessment remains necessary.

The third area concerns human-centered and cognitive capabilities. As AI applications become capable of performing a wider range of analytical and procedural tasks, research has continued to examine the place of creativity, critical thinking, communication, leadership, emotional intelligence, and ethical judgment in professional work. These competencies are often described as relatively durable because their application depends on interpersonal relationships, contextual understanding, and judgments that cannot always be specified through standardized procedures [93, 94]. Their importance does not imply that such activities are unaffected by AI. Rather, technological change can alter how these capabilities are exercised and how they interact with technical knowledge. Leadership, for example, involves decisions about the allocation of responsibilities, the interpretation of AI-supported recommendations, and the management of employees working with intelligent systems [95–97]. Ethical judgment is similarly relevant when organizations address algorithmic bias, transparency, accountability, and the handling of sensitive information [98]. The literature therefore points to the need to consider technical and human-centered competencies together, particularly where AI systems influence decisions with organizational or social consequences.

The fourth area concerns the organization of learning itself. Studies increasingly discuss workforce development in terms of continuing learning arrangements rather than training activities conducted only when a new system is introduced [99–101]. Internal corporate academies, microlearning platforms, mentoring arrangements, and opportunities for internal mobility represent different ways in which organizations may support the acquisition and application of new competencies [102]. These arrangements are relevant to knowledge management because skills acquired by individuals do not automatically become part of an organization's established practices. Employees also require opportunities to exchange experience, apply new knowledge, and learn from the outcomes of technology-related decisions. AI applications are increasingly discussed as components of training systems, including personalized learning pathways, performance feedback, and the identification of possible skill gaps [103–105]. In this setting, AI serves both as a subject of professional learning and as a technology used to organize learning activities [106]. However, the presence of such tools does not by itself establish their effectiveness. Their practical value depends on the suitability of the training content, employees' access to learning opportunities, managerial support, and the extent to which acquired competencies can be used in everyday work. Organizational culture and knowledge-sharing practices are therefore central to understanding how individual upskilling relates to wider organizational adaptation.

The fifth area concerns differences between sectors and organizational settings. Although many occupations face changing technological requirements, the competencies involved and the arrangements available for developing them are not uniform. In manufacturing, research commonly addresses automation, robotics, and Industry 4.0 systems, where employees may need to combine technical knowledge with an understanding of production processes and operational requirements [11, 107, 108]. Healthcare presents different concerns, including AI-assisted diagnosis, digital health records, professional responsibility, and the protection of patient information [109–111]. In education, attention is directed toward AI-supported teaching, curriculum development, and the preparation of educators and learners to use emerging technologies [112, 113]. Small and medium-sized enterprises face additional constraints associated with financial resources, technical expertise, and access to structured training opportunities [21, 114]. These differences suggest that workforce development cannot be adequately addressed through a single training model. Upskilling arrangements need to reflect the tasks performed, the technologies introduced, the regulatory environment, and the resources available within each organization [115, 116]. The five areas identified in this review thus provide a common basis for examining workforce transformation while allowing for substantial differences in how skills are developed and applied.

4.1 Practical Implications

The findings have implications for organizations, policymakers, educational institutions, and individuals. Across these groups, a recurring concern is how to maintain opportunities for learning while technological requirements continue to change. The review also suggests that effective responses need to account for differences in existing competencies, institutional resources, and the conditions under which newly acquired knowledge can be applied.

For organizations, upskilling is better understood as a continuing management responsibility than as a series of unrelated training events. Employee development can be connected more closely to the introduction of new technologies, changes in occupational tasks, and the identification of emerging competency requirements. Internal skills assessments, structured learning pathways, mentoring, and AI-supported training platforms may assist this process, provided that their use is appropriate to the organization's needs and resources. Equally important are opportunities for employees to share what they have learned and apply new competencies in their work. Managers therefore have a role not only in allocating training resources but also in establishing working conditions that support knowledge exchange, experimentation, and continued professional learning.

For policymakers, the findings point to the importance of examining workforce development alongside digital infrastructure, employment policy, and access to education. National and regional initiatives may include support for vocational training, workplace learning, and the development of basic AI literacy. Financial and institutional assistance may be particularly relevant to smaller enterprises and workers with limited access to formal training. Policy responses should also recognize that the availability of digital technologies does not necessarily correspond to equal opportunities for acquiring the competencies needed to use them. Differences in educational provision, employment conditions, and regional resources may influence participation in upskilling initiatives and should be considered when such programs are designed and evaluated.

For educational institutions, the findings raise questions about how curricula and professional learning opportunities respond to changes in occupational requirements. Universities, vocational institutions, and continuing education providers may need to review the place of digital literacy, AI-related technical knowledge, critical thinking, and ethical judgment within existing programs. Modular courses and opportunities for lifelong learning can provide alternatives to relying exclusively on conventional degree-based education. Cooperation with employers may also help institutions understand how competencies are applied in professional settings, although educational priorities should not be determined solely by immediate labor-market demand. A broader approach is needed to prepare learners for both current technological applications and future changes in knowledge-intensive work.

For individuals, the literature emphasizes the continuing importance of professional adaptability. Workers increasingly encounter situations in which established knowledge must be supplemented by new technical skills or applied in unfamiliar ways. Maintaining digital competence, understanding the uses and limitations of AI systems, and developing communication, judgment, and other human-centered capabilities are therefore relevant to long-term professional development. At the same time, responsibility for upskilling should not rest entirely with individual employees. Access to suitable learning opportunities, time for professional development, and support from employers and educational institutions also influence the ability to respond to technological change.

4.2 Research Gaps and Future Directions

Despite the growing literature on AI-related workforce development, several questions remain insufficiently examined. These gaps concern the duration and outcomes of upskilling initiatives, their organizational and economic consequences, differences across national and institutional settings, and the methods used to investigate workforce adaptation.

First, more longitudinal research is needed to examine how competencies develop and change after employees participate in AI-related training. Much of the literature discusses anticipated skill requirements, reports experiences at a particular point in time, or presents conceptual accounts of workforce transformation. Such approaches provide useful information about current concerns but offer limited evidence about whether newly acquired competencies are retained, transferred to everyday work, or remain relevant as technologies change. Studies following employees and organizations over longer periods could provide a clearer account of these processes and the conditions under which training is sustained.

Second, the economic and organizational outcomes of upskilling require closer examination. Although workforce development is frequently associated with productivity, innovation, and organizational competitiveness, these relationships are not always assessed using clearly defined outcome measures. Further research could examine training costs, changes in work performance, employee mobility, and the application of acquired knowledge within organizational processes. Attention should also be given to differences between short-term improvements in technical proficiency and longer-term changes in organizational capabilities. Such distinctions would make it easier to evaluate the practical consequences of different approaches to workforce development.

Third, greater attention is needed to differences across countries, regions, and institutional environments. The publication patterns reported in this review showed a relatively strong representation of China, the United States,

the United Kingdom, Australia, and India among records with available affiliation data. However, the geographical distribution of publications should not be treated as a direct measure of the extent or quality of upskilling practices in those countries. Research from lower-income regions and less frequently studied institutional settings remains important for understanding how access to infrastructure, educational provision, organizational resources, and employment conditions shapes workforce development. Comparative studies could examine whether approaches commonly discussed in technologically advanced settings are suitable under different economic and institutional conditions.

Fourth, methodological differences across the literature create difficulties when comparing findings and assessing the outcomes of particular interventions. Bibliometric studies, qualitative investigations, surveys, and conceptual analyses address different aspects of workforce transformation, but their findings are not always directly comparable. Further work could combine qualitative accounts of employee and organizational experience with longitudinal measurements of competency development and workplace outcomes. Where feasible, experimental or quasi-experimental designs could also be used to assess particular training interventions. Such approaches would help distinguish reported expectations about AI-related upskilling from evidence concerning its actual effects.

Future research would therefore be well served by longer-term evaluations, cross-country and cross-sector comparisons, and studies that examine how AI-supported learning arrangements operate in practice. Particular attention should be given to the relationship between individual skill acquisition, organizational knowledge-sharing practices, and measurable changes in work. These questions are relevant to understanding not only which competencies are required in the AI-driven knowledge economy but also how organizations and institutions can support their continued development.

4.3 Limitations

Several limitations should be considered when interpreting the findings of this review.

First, the analysis drew on 13,382 full-text records retrieved through EBSCO Discovery Service, and the size and composition of this dataset depended on the search settings applied. When more restrictive conditions were used in EBSCOhost, including the exclusion of related-word expanders, the restriction of searches to core indexing fields, and the removal of equivalent-subject expansion, the results were reduced to a more focused subset of 637 records. This substantial difference indicates that the retrieval strategy had a considerable influence on the literature represented in the analysis. The broader dataset was useful for describing publication patterns and frequently occurring terms, but it should not be regarded as equivalent to a set of 13,382 individually verified, eligible studies. In addition, restricting the review to English-language, peer-reviewed journal articles published between 2022 and 2025 may have excluded relevant work in other languages, earlier foundational studies, and research disseminated through other publication formats.

Second, the search strategy relied on a defined combination of terms concerning workforce development, AI, and the knowledge economy. Publications using alternative expressions, such as "digital capability building" or "workforce transformation," may not have been retrieved if they did not contain the required combinations of search terms. The results were also influenced by differences in how databases indexed subjects and recorded keywords. These factors limit the extent to which the retrieved records can be regarded as a complete representation of the relevant literature.

Finally, the study combined descriptive publication and keyword statistics with a discussion of recurring research themes. It did not conduct a quantitative meta-analysis or estimate the effects of particular upskilling interventions. Moreover, the frequency of a term in the retrieved records did not necessarily indicate its theoretical importance or establish a relationship between the concepts it represented. The findings should therefore be interpreted as a broad mapping and synthesis of recent research rather than as statistical evidence of the effectiveness of specific workforce development strategies. The five areas discussed in this review provide an organizing structure for understanding the literature, but their boundaries and relative importance require further examination using more narrowly defined and independently verified study samples.

5 Conclusions

This systematic literature review examined recent research on workforce upskilling in the AI-driven knowledge economy, drawing on peer-reviewed publications from 2022 to 2025. The findings indicate that the growing use of AI, automation, and data-intensive technologies is changing occupational requirements and the ways in which organizations approach employee development. Five interconnected dimensions were identified in the reviewed literature: foundational digital literacy, AI-specific technical competencies, human-centered capabilities, organizational learning systems, and sector-specific adaptation strategies. Together, these dimensions provide a basis for understanding workforce upskilling as a process involving both individual skill development and the organizational conditions under which knowledge is acquired, shared, and applied.

The review also points to a growing emphasis on continuous learning rather than isolated training activities designed to address immediate technical requirements. Digital and AI-related competencies are increasingly discussed alongside critical thinking, professional judgment, communication, and the ability to work with intelligent systems. At the organizational level, the findings suggest that acquiring new skills is only part of the challenge; opportunities to apply knowledge, exchange experience, and adapt established working practices are equally important. The differences observed across industries and organizational settings further indicate that upskilling strategies need to reflect specific occupational demands, institutional resources, and technological conditions. By bringing together these perspectives, the study clarifies the relationship between workforce competencies, organizational learning, and knowledge-based adaptation. It also identifies the need for further longitudinal and comparative research to establish how different approaches to upskilling affect employee development and organizational outcomes. In the AI-driven knowledge economy, the capacity to sustain learning and make effective use of human expertise remains a central consideration in organizational adaptation and innovation.

Author Contributions

Conceptualization, A.Š. and M.N.; methodology, A.Š.; software, M.N.; validation, A.Š. and M.N.; formal analysis, A.Š. and M.N.; investigation, M.N.; resources, A.Š.; data curation, A.Š. and M.N.; writing—original draft preparation, A.Š. and M.N.; writing—review and editing, A.Š. and M.N.; visualization, M.N.; supervision, M.N.; project administration, A.Š. and M.N.; funding acquisition, A.Š. and M.N. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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